Sample Papers

MTH501 Linear Algebra

Final Term Examination - February 2005

Time Allowed: 150 Minutes

Instructions

Please read the following instructions carefully before attempting any of the questions:

1. Attempt all questions. Marks are written adjacent to each question.
2. Do not ask any questions about the contents of this examination from anyone.
   - ***a.*** If you think that there is something wrong with any of the questions, attempt it to the best of your understanding.
   - ***b.*** If you believe that some essential piece of information is missing, make an appropriate assumption and use it to solve the problem.

****WARNING: Please note that Virtual University takes serious note of unfair means. Anyone found involved in cheating will get an `F` grade in this course.**

Total Marks: 65 **Total**
Questions: 12

Question No. 1 **Marks : 2**

If $\{v_1, v_2, v_3, \ldots, v_n\}$ be the orthogonal set of vectors then which statement(s) must be true.

I. $v_i \cdot v_j = 0$ for all.

II. $v_i \cdot v_i > 0$ for all *i and* $v_i \neq 0$.

III. Set$\{c_1 v_1, c_2 v_2, c_3 v_3, \ldots, c_n v_n\}$

1 I only.
2 I and II only.
3 II and III only
4 All of the three.

Question No. 2 **Marks : 10**

Find a least squares solution of the inconsistent system Ax =b where

$$A = \begin{bmatrix} 4 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix}, b = \begin{bmatrix} 2 \\ 0 \\ 11 \end{bmatrix}.$$

Question No. 3 **Marks : 5**

Find orthogonal projection of

$$y = \begin{bmatrix} 7 \\ 6 \end{bmatrix}$$ Onto $$u = \begin{bmatrix} 4 \\ 2 \end{bmatrix}.$$

Question No. 4 **Marks : 2**

If matrix A has zero as an eigenvalue then which statement(s) about A must be true.

I. Matrix A is not invertible.
II. Matrix A will also have an eigenvalue 2.
III. Matrix is diagonalizable.

1 II and III only.
2 I only.
3 II and III only.
4 All three.

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Question No. 5 **Marks : 2**

A Linear transformation $T: R^2 \to R^2$ is such that

$$T\begin{bmatrix} e_1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \& T\begin{bmatrix} e_2 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$$ then $$T\begin{bmatrix} 1 \\ -3 \end{bmatrix}$$ is,

1 $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$
2 $\begin{bmatrix} 1 \\ -3 \end{bmatrix}$
3 $\begin{bmatrix} 3 \\ 5 \end{bmatrix}$
4 $\begin{bmatrix} 1 \\ 4 \end{bmatrix}$

Question No. 6 **Marks : 10**

Determine whether the subset

$$\begin{bmatrix} 3 & 0 & -1 \\ 2 & 3 & 1 \\ -3 & 4 & 5 \end{bmatrix}$$?

If so find the corresponding eigenvector.

Question No. 11 **Marks : 10**

Show that the signal 2^k *and* $(-4)^k$ are the solution of the difference equation $y_{k+2} + 2y_{k+1} - 8y_k = 0$.

Find a basis $\{u_1, u_2, u_3\}$ for R^3 such that P is the change-of-coordinates matrix from $\{u_1, u_2, u_3\}$ to the basis $\{v_1, v_2, v_3\}$.

Question No. 12 **Marks : 5**

Find the vector x such that we have

$$[x]_B = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$$

(Coordinate matrix of x with the basis B), where

$$B = \left\{ \begin{bmatrix} 3 \\ -5 \end{bmatrix}, \begin{bmatrix} -4 \\ 6 \end{bmatrix} \right\}.$$